A Resource-Efficient Hardware Accelerator for Lossless Image Compression Based on the Walsh–Hadamard Transform

1st Marco Zolla Salazar
Maestría en Ingeniería Electrónica
Instituto Tecnológico de Costa Rica
Costa Rica
mzolla@estudiantec.cr

2nd Luis Diego Martinez Jimenez
Maestría en Ingeniería Electrónica
Instituto Tecnológico de Costa Rica
Costa Rica
diego_mj22@estudantec.cr

3rd Simón Fallas Villalobos
Maestría en Ingeniería Electrónica
Instituto Tecnológico de Costa Rica
Costa Rica
simonfallasv@estudiantec.cr

4th Marcelo Mata Solano
Maestría en Ingeniería Electrónica
Instituto Tecnológico de Costa Rica
Costa Rica
m.mata.3@estudiantec.cr

5th Brandon Fabricio Rivera Hidalgo
Maestría en Ingeniería Electrónica
Instituto Tecnológico de Costa Rica
Costa Rica
b.rivera.2@estudiantec.cr

***Abstract*—Placeholder abstract. To be written for the final article (max. 250 words).**

***Index Terms*—Discrete transforms, Image coding, Data compression, Field programmable gate arrays, Hardware acceleration, High level synthesis**

I. Introduction

Modern high-frame-rate camera systems, such as those deployed in intelligent vehicles and satellite imaging, generate massive volumes of data that must be processed and stored with minimal latency. In many of these edge-computing scenarios, lossless compression is essential because the uncertainty introduced by quantization noise in lossy systems is often inadmissible for maintaining system robustness or meeting legal requirements [7]. Implementing these codecs in real-time at the edge is a critical challenge, as embedded video processors must handle high-speed transform computations while operating under strict limitations regarding hardware area, power consumption, and thermal dissipation [11].

Transform methods are a fundamental component of image compression because they decorrelate spatial data before encoding, concentrating information into a smaller number of coefficients. Among these, the Walsh-Hadamard Transform (WHT) is particularly attractive for hardware implementations due to its significant computational advantages over sinusoidal transforms like the DFT or DCT. Because the WHT kernel matrix consists entirely of integer values of +1 and -1, the transform can be implemented very efficiently using only additions and subtractions, thereby eliminating the need for costly and time-consuming multipliers [1], [4].

Despite its advantages, state-of-the-art literature reveals limitations in existing solutions, which are frequently optimized for lossy applications or general neural network acceleration rather than resource-efficient lossless compression. For example, some high-throughput WHT-based architectures are designed for signal processing but rely on 32-bit floating-point arithmetic, which is not ideal for exact, low-resource lossless operation [1]. Conversely, other lossless approaches, such as those based on the JPEG-LS standard, provide competitive compression but often do not leverage the unique hardware-friendly properties of the WHT [7].

In this paper, we propose a resource-efficient integer/reversible WHT accelerator specifically designed for low-latency lossless compression. Our main contributions are:

- An optimized (integer/reversible) WHT architecture that ensures mathematically exact data reconstruction for lossless applications.
- The development of a flexible design modeled at a high level (SystemC/HLS), allowing for rapid design space exploration and the generation of parallel-pipeline layouts.
- A comprehensive evaluation on FPGA hardware to verify performance metrics, demonstrating improvements in throughput and area efficiency compared to existing implementations.

The remainder of this paper presents the *Background and Related Work*, providing a review of the WHT mathematical foundations and current hardware implementation strategies. The *Proposed Solution / Architecture*, describing our high-level modeling approach and the specific hardware structures of the integer WHT accelerator. *Results and Analysis*, detailing the experimental setup and providing a comparative analysis of our results on FPGA. And finally, the *Conclusions and Future Work*, which summarizes our findings and outlines potential avenues for further optimization in real-time edge compression.

II. Background and Related Work

The Walsh–Hadamard transform (WHT) decomposes a signal into Walsh basis functions whose values are restricted to ± 1. For a signal of 2^n samples the transform is a multiplication by the $2^n \times 2^n$ Hadamard matrix, and since its entries are ± 1 it can be evaluated with additions and subtractions alone. The fast WHT (FWHT) lowers the arithmetic cost from $O(N^2)$ to $O(N \log_2 N)$ through Cooley–Tukey-style factorizations [1], and its parallel-pipeline hardware implementations follow processor structures first developed for the fast Fourier transform [14].

Exact reconstruction is what keeps the WHT from being lossless. Its inverse normalizes by the transform size N, a division that integer arithmetic cannot represent exactly, so a direct WHT loses information over a round trip through the transform and its inverse. Integer, reversible formulations remove this limitation. The unified lossless WHT (ULWHT) recovers the original data exactly while lowering the per-coefficient entropy of natural images [2], and a lossless two-dimensional WHT has also been derived [3]. For hardware there is a caveat: the one-dimensional lossless WHT still contains multiplications, and making it fully multiplier-free needs a more elaborate construction [3].

Two closely related solutions provide reproducible results with public code, and between them they cover both sides of the problem. One is an FPGA WHT accelerator: a configurable parallel-pipeline FWHT produced by high-level synthesis, with its HLS source openly released [1]. The other is a parallel lossless compressor for raw Bayer images on an FPGA-based high-speed camera that sustains 40.1 Gbit/s at 320 MHz on a Zynq UltraScale+ device, using adaptive Golomb–Rice coding and a public reference implementation [5]. One supplies the transform, the other a low-latency lossless pipeline; bringing the two together is the open problem.

Five further works address parts of the problem. In hardware, a multiplier-free WHT running at 162 MHz takes the place of the Photo Core Transform in JPEG XR [4], an FPGA JPEG 2000 accelerator performs on-board lossless compression [6], and an ANS-based coder reduces the complexity of JPEG-LS [7]. The integer lossless WHT formulations achieve bit-exact compression but stop at the algorithmic level, with no hardware realization [2], [3]. Earlier WHT processors based on distributed arithmetic [11], the sequency-ordered complex Hadamard transform [12], and a radix-2 Walsh–Hadamard–Fourier pipeline [13] were built for high throughput and small area, drawing on the low complexity of the ± 1 kernel [8].

The WHT also appears outside compression, in image watermarking [15], ultrasonic coded excitation [9], and the simulation of quantum algorithms [10]. Across this literature we found no hardware accelerator from the past decade that performs lossless, WHT-based compression of high-frame-rate image streams. The reproducible accelerators cover either the transform [1] or the lossless coding [5], and the lossless integer WHT has stayed at the algorithmic level [2], [3]. Closing this gap is the aim of the present work.

III. Proposed Solution

A. Study of Other Solutions in the State of the Art

An analysis of recent literature over the past ten years reveals that implementations oriented toward image compression bifurcate into approaches that partially cover the problem but lack an efficient architectural integration specifically oriented toward a multiplier-free design for lossless video streams.

1) Hardware Accelerators for Transforms (Lossy Approach): Hardware accelerators for transforms oriented toward lossy approaches have demonstrated remarkable efficiency in the real-time domain. These architectures achieve high data processing throughput by utilizing parallel and pipelined logic structures (*parallel-pipeline*) [1], also proving the physical feasibility of operating at high clock frequencies, such as 162 MHz, completely omitting advanced DSP logic blocks [4]. Despite these strengths, these solutions present severe limitations when considered for lossless environments, as they are optimized for generic processing or traditional lossy schemes relying on 32-bit floating-point arithmetic [1]. Furthermore, using the conventional inverse transform introduces division operations for normalization that integer-based systems cannot represent exactly, injecting quantization noise that destroys the possibility of exact data reconstruction [2]. Consequently, while these designs represent a valuable metric baseline for evaluating hardware cost in terms of area and speed, they are unfeasible for strictly lossless compression applications.

2) Lossless Hardware Compressors (Complete Pipelines): On the other hand, developments focused on complete lossless compression pipelines offer robust, low-latency alternatives essential for edge computing. Solutions such as parallel compressors for raw Bayer images implemented on FPGAs [5] and structural optimizations a pplied to the JPEG-LS standard under advanced coding architectures [7] can sustain massive data transfer rates in real-time. However, the main deficiency of this approach is that optimization efforts are directly concentrated on the entropy coding and packaging stages, omitting or completely replacing the spatial decorrelation phase based on discrete transforms such as the WHT [7]. This omission makes them highly viable for general-purpose compression at the edge, but unfeasible if the architectural design goal is to maximize datapath efficiency by exploiting a multiplier-free transformation core composed exclusively of $+1$ and -1 coefficients [4].

3) Reversible WHT Algorithmic Formulations: Finally, proposals focused on the theoretical formulations of the reversible transform resolve the mathematical limitations of information loss in the inverse process. Mathematical models such as the unified lossless Walsh-Hadamard transform (ULWHT) and its two-dimensional derivations formally demonstrate that it is viable to recover data exactly bit-by-bit, significantly reducing the entropy per coefficient without resorting to complex multiplication operations [2], [3]. The major weakness of this approach lies in its purely abstract nature, given that these formulations have remained strictly in the algorithmic domain and lack a tangible realization or microarchitectural mapping in hardware for embedded systems [2]. Thus, although they are completely viable as the numerical reference model for the system, they are unfeasible for direct physical deployment without a profound process of architectural exploration and synthesis.

B. Observed Challenges

Based on the critical study of the state of the art, the fundamental architectural challenges that justify the contribution of this research are identified and outlined. The first challenge consists of properly delimiting the functional scope of the accelerator. Since the WHT transform acts solely as a spatial decorrelator and not as an autonomous codec, it is imperative to narrow the architecture to a dedicated transform core accelerator, structured in such a way that it can be seamlessly integrated into a larger lossless pipeline without diverting logic efforts toward complex entropy coding. The second technical challenge lies in performing a direct mapping from the abstract algorithmic domain to a real physical architecture with zero use of DSP blocks. Although the literature describes theoretical reversible and multiplier-free versions [3], their implementation through high-level synthesis (SystemC/HLS) requires optimizing subexpression reuse and designing aggressive pipelining to mitigate FPGA resource consumption.

As a third challenge, there is a need to strongly justify the choice of the WHT over competing integer transforms such as the DCT or wavelets. Because the WHT exhibits inferior performance in lossy compression quality [5], the defense of the project must be empirically grounded in its superiority regarding logic area cost and maximum operating frequency in a lossless environment. The fourth challenge focuses on ensuring real-time operational performance for high-frame-rate video streams, which forces the design of a datapath with a highly optimized transaction-level data flow that minimizes stall cycles under the strict thermal and power constraints of embedded systems [11]. Finally, the fifth challenge corresponds to the verification of bit-exact reproducibility. To ensure an identical reconstruction of the original image without compromising fidelity, a robust reference model will be developed in simulation environments such as MATLAB and Simulink, which will serve as the definitive benchmark to validate the behavior of the SystemC transactional model and the synthesized hardware through rigorous testing.

C. Scope and Design Philosophy

Based on the analysis presented in Section II, there is currently no FPGA-based hardware accelerator that implements a lossless Walsh-Hadamard Transform (WHT) while maintaining optimized resource utilization and real-time operation. Therefore, this work focuses on the development of an accelerator dedicated exclusively to the execution of the transform. The proposed system performs an integer and reversible Walsh-Hadamard Transform on pixel blocks obtained from a high-speed video stream. The output consists of transform coefficients that can subsequently be processed by an entropy coding stage, such as Golomb-Rice or ANS coding. This design boundary allows the evaluation of only the hardware associated with the transform itself, quantifying its efficiency in terms of resource utilization and performance without being influenced by the compression ratio of a complete codec.

D. High-Level Architecture

The proposed architecture consists of four main functional blocks, as illustrated in Fig. 1.

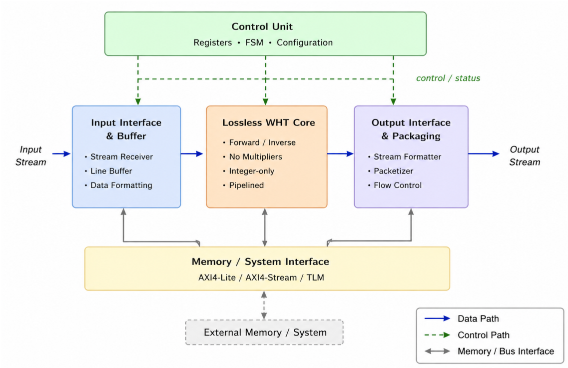

Fig. 1. High-level architecture of the proposed lossless WHT accelerator.

1) Input Buffer and Interface (IF): This block receives pixels from the camera or memory subsystem and organizes them into $N \times N$ blocks, where $N \in \{8, 16, 32\}$. The final block size will be determined during the design space exploration phase, as it directly affects the trade-off between decorrelation capability, memory requirements, and processing latency.

2) Lossless WHT Core: The computational core of the accelerator was designed in Vitis HLS utilizing a fully parallel, multiplier-free architecture. To satisfy the requirements of a lossless environment (zero quantization noise) while keeping DSP utilization at zero, the standard fast Walsh-Hadamard butterfly was replaced by an integer lifting-based scheme (S-Transform) [2].

The fundamental spatial decorrelation step between two pixels a and b is implemented purely through high-frequency differences ($d = a - b$) and low-frequency shifted averages ($s = a - (d \gg 1)$). This specific subexpression routing is synthesized by the HLS compiler completely into LUTs and Flip-Flops, bypassing any DSP slice instantiation.

To maximize datapath throughput for real-time video, the transform operations for a block size of $N = 8$ were flattened into three pipeline stages. We enforce maximal hardware concurrency by applying the `#pragma HLS PIPELINE II=1` and `#pragma HLS UNROLL` directives. Additionally, variables were strictly constrained using HLS arbitrary precision integers (`ap_int<16>`), which prevents the datapath from unnecessarily routing 32-bit standard C++ integer bounds, ensuring a highly resource-efficient mapping on the Kria KV260 architecture. The complete HLS source code for the transform core is publicly available in our repository [16].

3) Control Unit: The control unit manages the overall operation of the accelerator. Although it does not directly perform the WHT computation, it coordinates when pixel blocks enter the system, when temporary data are stored in registers, and when results are forwarded to the next pipeline stage. Furthermore, it synchronizes processing to prevent data conflicts and enables architectural parallelism through control signals such as enable, valid, and ready.

4) Memory Interface (AXI/TLM): This module manages data transfers between the accelerator and the external memory subsystem. The use of AXI/TLM interfaces facilitates integration into FPGA-SoC platforms such as the Kria KV260 and Alveo U55C, allowing the accelerator to be incorporated into larger embedded processing systems.

E. Challenge–Solution Mapping

The proposed architecture aims to address the main challenges identified during the literature review. Table I summarizes how each challenge is considered within the proposed solution.

TABLE I
CHALLENGE–SOLUTION MAPPING

Challenge	**Proposed Solution**
D1 – Codec Scope	This work focuses only on the WHT stage. Entropy coding (e.g., Golomb-Rice, ANS) is considered a later processing stage, allowing the transform hardware to be evaluated independently.
D2 – DSP-Free FPGA Implementation	The WHT core uses only additions, subtractions, and arithmetic right shifts based on the lifting construction from [2], minimizing DSP usage and mapping entirely to LUTs and Flip-Flops.
D3 – WHT vs. DCT/Wavelets	The proposed architecture will be compared against integer DCT implementations using hardware resource metrics (LUT/FF/DSP), justifying the WHT on hardware efficiency grounds.
D4 – Real-Time Throughput	A fully pipelined datapath is proposed to support real-time video processing, sustaining one block per N clock cycles.
D5 – Bit-Exact Lossless Reconstruction	The reversible lifting-based formulation guarantees exact reconstruction. A Python/C reference model will be used to perform sample-by-sample comparison against hardware output.

Overall, the proposed architecture provides a feasible FPGA implementation of a lossless Walsh-Hadamard Transform while addressing the main limitations identified in previous studies.

IV. RESULTS AND ANALYSIS

At this preliminary stage we report the functional verification of the transform core and its effect on the coding cost of the data. Resource and timing figures from logic synthesis (LUT, FF, DSP utilization and maximum operating frequency) will complete the analysis.

A. Experimental Setup

A bit-accurate reference model of the transform was implemented in SystemC and used as the verification oracle. The model reproduces the integer lifting butterfly of the HLS core and adds the inverse transform, which the synthesizable core does not implement. All experiments are reproducible from the public repository [16]: the reference model, the test benches, and the datasets are generated deterministically from a fixed seed. The evaluation uses five 8-bit grayscale 256×256 images spanning the range of spatial correlation: a constant field, a linear ramp, a piecewise-constant image with sharp edges, a $1/f$ surface with spatial correlation similar to natural images, and white noise as a control. The transform is applied over one-dimensional $N = 8$ strips, matching the hardware; a separable two-dimensional variant (row then column) is also evaluated as a software-only extension.

B. Functional Verification

Two properties were checked. First, source-level equivalence (at the C/C++ level, before synthesis) between the reference model and the HLS core: over a fixed test vector and 10^5 random 8-bit blocks both produce identical coefficients, with no divergence. Second, losslessness: the inverse of the forward transform recovers the input exactly over 4×10^5 blocks, including the full signed 16-bit range, which confirms that the construction is exactly reversible. A separate check compares the 16-bit datapath against an extended-precision (wide-integer) reference that cannot wrap and finds no overflow for 8-bit input, so 16 bits are sufficient for the target dynamic range.

C. Entropy Reduction

Since the accelerator implements the transform stage only, its benefit is measured as the reduction in zeroth-order entropy (bits per symbol) between the pixels and the transform coefficients. Entropy is the lower bound on the rate of any zeroth-order lossless coder, so a lower value indicates more compressible data. The length-8 transform maps each block to one DC (mean) coefficient and seven detail coefficients of increasing sequency, which we treat as eight frequency bands. We report a pooled estimate (all coefficients in one histogram) and a per-band estimate (one entropy per frequency position, averaged), the latter being closer to the cost of a coder that models each band separately. Table II lists the results.

TABLE II
ZEROTH-ORDER ENTROPY (BITS/SYMBOL): PIXELS VS. WHT COEFFICIENTS.

		1D (N=8, HW)		2D separable	
Dataset	H_{pix}	pool	band	pool	band
Flat	0.00	0.54	0.00	0.12	0.00
Ramp	8.00	2.63	0.63	0.54	0.08
Edges	2.00	0.79	0.25	0.15	0.03
Natural	7.30	6.55	5.81	5.86	5.31
Noise	8.00	8.75	8.20	8.77	7.84

The reduction tracks the spatial correlation of the input. Smooth images (ramp, edges) concentrate almost all information in the DC band and drive the detail bands to zero entropy, giving per-band reductions of $87–92\%$. For the natural-like image the per-band reduction is 20% in one dimension and 27% for the separable two-dimensional transform. White noise, having no spatial correlation, is not improved by the transform and serves as a control; its coefficient entropy even exceeds the 8 bits of the input because the coefficients span a wider dynamic range than the pixels, which is also why the datapath is 16 bits wide. The gap between the one- and two-dimensional results motivates a two-dimensional core as future work.

Declaration on the Use of Generative AI

During the preparation of this work, the authors used an AI-based assistant to consult concepts on the Walsh–Hadamard transform and lossless coding, to survey the state of the art and verify each cited source against its full text (author names, numerical results, and DOIs), to format the references in IEEE style and search for the index terms from the IEEE Thesaurus, and to review the manuscript critically for factual accuracy against the cited sources.

References

[1] A. Manjarres Garcia, C. Osorio Quero, J. Rangel-Magdaleno, J. Martinez-Carranza, and D. Durini Romero, "Parallel-pipeline fast Walsh-Hadamard transform implementation using HLS," in *Proc. Int. Conf. Field-Programmable Technology (ICFPT)*, 2021, doi: 10.1109/ICFPT52863.2021.9609874.

[2] H.-Y. Jung, R. Prost, and T.-Y. Choi, "A unified mathematical form of the Walsh–Hadamard transform for lossless image data compression," *Signal Process.*, vol. 63, pp. 35–43, 1997, doi: 10.1016/S0165-1684(97)00138-2.

[3] K. Komatsu and K. Sezaki, "Lossless 2D discrete Walsh-Hadamard transform," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2001, doi: 10.1109/ICASSP.2001.941320.

[4] S. Hafizullah, M. S. S. V. S. Manideep, V. Sharma, P. K. Nath, A. Naugarhiya, and S. Verma, "An efficient hardware implementation of Walsh Hadamard transform for JPEG XR," in *Proc. 15th IEEE India Council Int. Conf. (INDICON)*, 2018, doi: 10.1109/INDICON45594.2018.8987180.

[5] Ž. Regoršek, A. Gorkič, and A. Trost, "Parallel lossless compression of raw Bayer images on FPGA-based high-speed camera," *Sensors*, vol. 24, no. 20, art. 6632, 2024, doi: 10.3390/s24206632.

[6] R. Ghodhbani, T. Saidani, L. Horrigue, A. M. Algarni, and M. Alshammari, "An FPGA accelerator for real-time hyperspectral image compression based on JPEG2000 standard," *Eng. Technol. Appl. Sci. Res.*, vol. 14, no. 2, pp. 13118–13123, 2024, doi: 10.48084/etasr.6853.

[7] T. Alonso, G. Sutter, and J. E. López de Vergara, "LOCO-ANS: An optimization of JPEG-LS using an efficient and low-complexity coder based on ANS," *IEEE Access*, vol. 9, pp. 106606–106626, 2021, doi: 10.1109/ACCESS.2021.3100747.

[8] D. Nayak, K. Ray, T. Kar, and C. Kwan, "Walsh–Hadamard kernel feature-based image compression using DCT with bi-level quantization," *Computers*, vol. 11, no. 7, art. 110, 2022, doi: 10.3390/computers11070110.

[9] C. Weng, X. Gu, and H. Jin, "Coded excitation for ultrasonic testing: A review," *Sensors*, vol. 24, no. 7, art. 2167, 2024, doi: 10.3390/s24072167.

[10] A. Kobori, R. Takahashi, and M. Nakanishi, "A hardware architecture for the Walsh–Hadamard transform toward fast simulation of quantum algorithms," *CCF Trans. High Perform. Comput.*, vol. 2, pp. 211–220, 2020, doi: 10.1007/s42514-020-00028-7.

[11] A. Amira and S. Chandrasekaran, "Power modeling and efficient FPGA implementation of FHT for signal processing," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 15, no. 3, pp. 286–295, 2007, doi: 10.1109/TVLSI.2007.893606.

[12] G. Bi, A. Aung, and B. P. Ng, "Pipelined hardware structure for sequency-ordered complex Hadamard transform," *IEEE Signal Process. Lett.*, vol. 15, pp. 401–404, 2008, doi: 10.1109/LSP.2008.922515.

[13] J. Liu, Q. Xing, X. Yin, X. Mao, and F. Yu, "Pipelined architecture for a radix-2 fast Walsh–Hadamard–Fourier transform algorithm," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 62, no. 11, pp. 1083–1087, 2015, doi: 10.1109/TCSII.2015.2456371.

[14] E. Wold and A. Despain, "Pipeline and parallel-pipeline FFT processors for VLSI implementations," *IEEE Trans. Comput.*, vol. C-33, no. 5, pp. 414–426, 1984, doi: 10.1109/TC.1984.1676458.

[15] A. T. Ho, J. Shen, A. K. Chow, and J. Woon, "Robust digital image-in-image watermarking algorithm using the fast Hadamard transform," in *Proc. Int. Symp. Circuits Syst. (ISCAS)*, 2003, doi: 10.1109/ISCAS.2003.1205147.

[16] MP-6160 Grupo 5, "WHT Lossless Accelerator Repository." [Online]. Available: https://github.com/MP6160-Grupo-5/MP6160_grupo_5